

ADAPTING SEGMENT ANYTHING 2 FOR DIABETIC RETINOPATHY LESION SEGMENTATION

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ABSTRACT

Automated segmentation of diabetic retinopathy (DR) lesions in fundus images plays a crucial role in computer-aided diagnosis. While existing research has explored lesion segmentation extensively, many approaches rely on traditional, smaller backbone networks for feature extraction. In contrast, foundation segmentation models, benefiting from pre-training on vast, high-quality segmentation datasets, offer more efficient feature extraction capabilities. Therefore, this paper proposes to adapt the Segment Anything Model 2 (SAM2) to enhance DR lesion segmentation. Specifically, we freeze the parameters of the SAM2 Hierarchical encoder and leverage adapter modules for efficient fine-tuning. The decoder employs a multi-scale block design, mirroring the encoder's architecture, to achieve efficient segmentation. Experimental validation on the publicly available IDRiD and DDR datasets demonstrates the effectiveness of our proposed network.

Index Terms— Segment Anything Model 2, Medical Image Segmentation, Retinopathy Lesion Segmentation, Deep Learning

1. INTRODUCTION

Automated medical image segmentation [12, 16] plays a crucial role in computer-aided diagnosis. Lesion segmentation in fundus images can serve as a helpful tool in the early detection and monitoring of various retinal diseases. However, this task is challenging due to some intrinsic properties of the lesion, as shown in Figure 1. Specifically, lesions differ in their location, shape, and size. Also, many lesions occupy only a small portion of the overall fundus image, causing a serious class imbalance. In addition, the boundary between many lesions and normal areas is not clear, making it a challenge to accurately identify the edge. Therefore, the accurate identification of lesions is always a meaningful and challenging problem.

With the development of deep learning technology, many methods for fundus lesion segmentation [4, 17, 8, 3, 7, 15] have emerged. L-Seg [4] designs a multi-scale feature fusion method to better handle small lesion regions. HACDR-

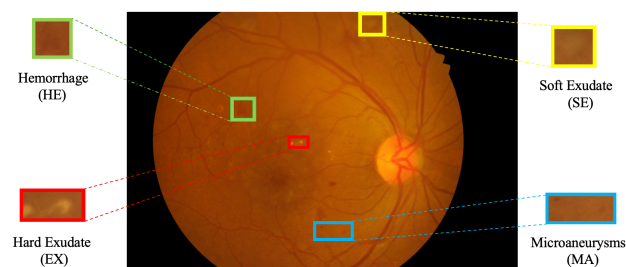


Fig. 1. An example of diabetic retinopathy lesions image from IDRiD [10].

Net [17] leverages heterogeneous cross-convolution, heterogeneous modulated deformable convolution, and optional near-far-aware convolution to conquer the challenges of various lesion shapes and sizes. WSRFNet [8] refines multi-scale features using a recurrent feedback mechanism. More recently, there has also been some work attempting specialized feature encoders to improve lesion segmentation performance. For example, SEBNet [3] designs a multi-step scale embedding block-based encoder, which comprises two branches that operate on different spatial resolutions to enhance multi-resolution interaction. However, due to the limited number of network parameters and lack of high-quality segmentation pre-training, the ability of these methods to extract robust features is still lacking.

Foundation segmentation models, such as the Segment Anything Model (SAM) [5], offer the possibility of achieving more accurate image segmentation. Although SAM's zero-shot performance on medical images is poor, there has been a lot of work [1, 2] demonstrating that parameter-efficient fine-tuning techniques can effectively mine powerful representations in the foundation model to significantly improve performance on downstream tasks. Inspired by this success, in this paper, we propose to adapt the Segment Anything Model 2 (SAM2) [11] for lesion segmentation. Specifically, we froze the parameters of the SAM2 encoder and inserted a learnable adapter block into it, which greatly reduced the memory overhead required for fine-tuning. Further, we abandoned SAM2's original decoder and instead used multi-scale blocks to build

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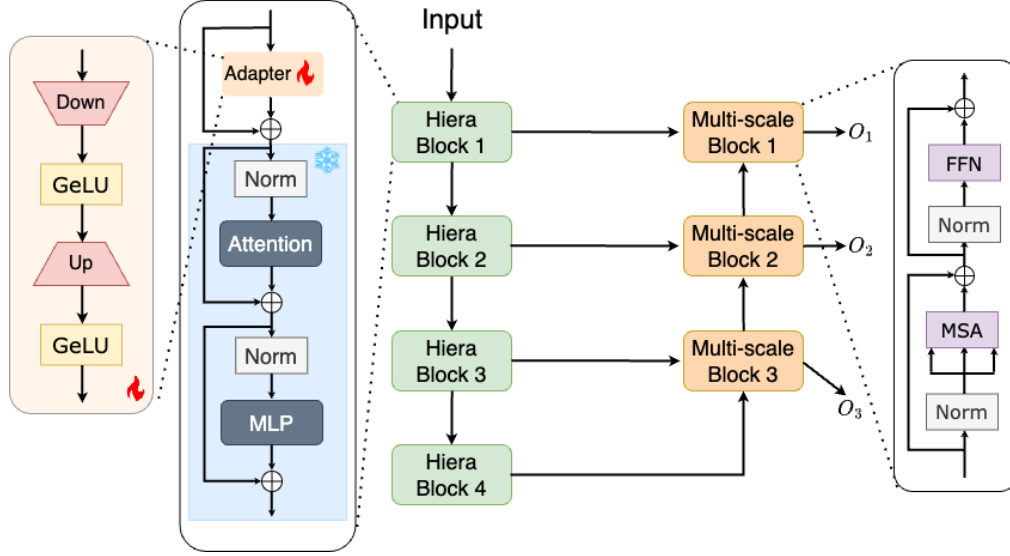


Fig. 2. The overview of our proposed network.

a U-Net style decoder, which has been widely proven to be effective in medical image segmentation. We conducted experiments on two publicly available datasets, IDRiD [10] and DDR [6], and the experimental results demonstrate the validity of our approach.

2. METHOD

Fig. 2 illustrates the overview of our framework, which is mainly based on the SAM2 [11] model. For the encoder, we apply the Hiera [13] backbone of SAM2, while the decoder consists of a series of multi-scale blocks.

2.1. The Adapted SAM2 Encoder

The easiest way to transfer SAM2 to downstream tasks is to fine-tune all of its parameters directly. However, due to the relatively large number of parameters in the foundation model, direct full fine-tuning would result in a large memory consumption, which is inefficient. Parametric efficient fine-tuning techniques aim to fine-tune only a small part of a model’s parameters to achieve results comparable to full fine-tuning, and have been proven useful in many tasks. Inspired by this, in our network, we also froze all parameters of the Hiera encoder in SAM2 and inserted learnable adapter blocks for more efficient transfer learning. Specifically, the adapter block leverages a lightweight encoder-decoder design, which begins with a linear layer to squeeze dimensions, followed by a GeLU activation function for non-linearity, a linear layer to expand dimensions, and a final GeLU operation.

2.2. The Multi-Scale Decoder

The original SAM2 decoder uses a lightweight, promptable transformer architecture. This architecture is more suitable for video segmentation and tracking tasks, but it is suboptimal in lesion segmentation. On the one hand, lesion segmentation is an end-to-end automatic process, and it is unrealistic to rely on medical experts to give manual prompts during inference. Therefore, the newly designed decoder should be designed to be prompt-free. On the other hand, considering the large differences between the shapes, sizes, and positions of the lesions, simply using a lightweight decoder is not enough to adequately decode the rich lesion-aware features provided by the encoder. A careful design of hierarchical decoders is needed to better identify various lesions.

Specifically, our multi-scale block is mainly adapted from the design in [18], which is composed of a Multi-scale Attention (MSA) and a Feed-Forward Network (FFN). The MSA is formulated as follows:

$$\begin{aligned} Q &= f_q(x_{in}), K = V = f_{kv}(\text{Pool}(x_{in}, R)), \\ \text{Att} &= \text{Softmax}(Q \cdot K^T / \sqrt{d_k}), x_{out} = \text{Att} \cdot V, \end{aligned} \quad (1)$$

where x_{in} denotes the input feature, Q , K , V denotes the query, key, and value matrix that is derived from x_{in} , Pool denotes leverage of the average pooling operation to reduce the spatial resolution to improve efficiency, and R denotes the pooling rate. Similar to many transformer designs, our FFN uses a simple bottleneck structure, as follows:

$$\text{FFN}(x_{in}) = \text{Linear}((\text{DW}(\text{Linear}(x_{in}))), \quad (2)$$

where DW indicates the depth-wise convolution. As the MSA and FFN are connected sequentially, the whole process of our

Table 1. Model performance on the IDRiD [10] and DDR [6] datasets.

Method	IDRiD [10]			DDR [6]		
	mDice	AUC-ROC	AUC-PR	mDice	AUC-ROC	AUC-PR
U-Net [12]	36.66	89.34	34.68	24.59	91.36	26.78
FCT [14]	42.96	93.49	43.09	23.33	90.53	25.01
Vanilla SAM [5]	1.35	45.35	0.61	0.51	34.56	0.28
MedSAM [9]	25.06	87.27	26.57	17.97	89.83	19.30
SAMed [19]	35.18	88.96	35.37	28.32	87.70	30.03
SAM-Adapter [1]	34.42	88.23	34.59	29.11	88.82	29.90
Ours	51.39	96.01	51.62	39.42	94.85	39.76

multi-scale block is formulated as:

$$\begin{aligned} z &= \text{MSA}(\text{LN}(x_{in})) + x_{in}, \\ x_{out} &= \text{FFN}(\text{LN}(z)) + z, \end{aligned} \quad (3)$$

where z is the intermediate features, and LN is a layer normalization. Overall, our decoder consists of multiple multi-scale blocks that recover lesion details bottom-up in a U-shape style for accurate segmentation. In addition, in the intermediate decoding stage, we also output side results to achieve deep supervision.

2.3. Loss Function

We employ a combination of weighted IoU loss and weighted binary cross-entropy loss [16] to supervise each lesion category:

$$\mathcal{L} = \mathcal{L}_{\text{IoU}}^w + \mathcal{L}_{\text{BCE}}^w. \quad (4)$$

Furthermore, to facilitate model convergence and increase robustness, we apply deep supervision to the different side outputs of the model.

3. EXPERIMENTS

3.1. Datasets

To assess the performance of our approach, we conduct experiments on two public retinal lesion segmentation datasets, IDRiD [10] and DDR [6]. Both the two datasets include four categories of Diabetic Retinopathy (DR) lesion regions (MA, HE, EX, and SE) derived from color fundus images. The IDRiD dataset comprises 81 high-quality images, all at a uniform resolution of 4288×2848 , with 54 images designated for training and 27 for testing. The DDR dataset contains 757 color fundus images, divided into 383 for training, 149 for validation, and 225 for testing, with resolutions ranging from 1088×1920 to 3456×5184 .

3.2. Implementation Details

Our network is implemented using PyTorch, and all experiments are carried out on two NVIDIA RTX 4090 GPUs. We

train the DDR dataset for 200 epochs and the IDRiD dataset for 500 epochs. Input images are resized to 1024×1024 during both training and inference. The AdamW optimizer is employed with a learning rate set to 0.0001. For evaluation, we utilize the following metrics: mDice, AUC-ROC, and AUC-PR.

3.3. Result Analysis

To validate the performance of our network for diabetic retinopathy lesion segmentation, we compare its performance against the following methods: U-Net [12], FCT [14], vanilla SAM [5], MedSAM [9], SAMed [19] and SAM-Adapter [1]. The results are shown in Table 1. It can be found that our method is superior to other approaches across both datasets. For the IDRiD dataset, our method achieves results of 51.39%, 96.01%, and 51.62% in mDice, AUC-ROC, and AUC-PR respectively. For the DDR dataset, Our method achieves results of 39.42%, 94.85%, and 39.76% in mDice, AUC-ROC, and AUC-PR respectively. These results highlight the superiority of our method compared to other SAM-based methods.

4. CONCLUSION

In this paper, we propose a novel segmentation framework that adapts the Segment Anything Model 2 for diabetic retinopathy lesion segmentation. By utilizing the high-quality feature in the SAM2 backbone, our approach achieves better segmentation effectiveness. In future work, we plan to research how to improve the adapter and decoder design to achieve more efficient and accurate segmentation.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by [10, 6]. Ethical approval was not required as confirmed by the license attached with the open access data.

6. ACKNOWLEDGMENTS

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